FLEET TEAM DIGEST

737MAX-FTD-32-23001

Issue Title : Thermal Oxidation Damage - Carbon Brakes

Airplane Model	ATA	Minor Model(s)
737MAX	3241-41	-10, -7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
07/12/2023	11/22/2023	11/14/2023

Estimated Completion Date	Next Update Date

Revision Description

Added additional information regarding the adjusted autobrake option that is only available on the MAX fleet.

Description

Boeing has observed an increasing number of events where carbon brake assemblies have experienced thermal oxidation damage. This Fleet Team Digest (FTD) article is intended to provide information regarding this phenomenon and how to manage brake energy levels in order to reduce the likelihood of thermal oxidation damage.

Status

Thermal oxidation occurs when carbon is exposed to high temperatures (braking energy) followed by exposure to oxygen contained in the air. The excess heat in the brake assembly causes a chemical reaction where the carbon atoms combine with oxygen atoms forming carbon dioxide. Because carbon dioxide is a gas, oxidized carbon will lose mass and become structurally weakened. Thermal oxidation rate is a function of the duration at which the carbon is exposed to elevated temperatures.

In order for each brake supplier to protect the carbon brake from this phenomenon, each supplier applies their proprietary Oxidation Protection System (OPS) for the carbon. This coating is designed to withstand high temperatures. However, if the design temperature of the OPS is exceeded, there are cases where high brake energies can cause the coating to become damaged and/or deteriorated. Once the coating becomes damaged and/or deteriorated, normal operating brake temperatures can cause thermal oxidation damage. The hottest location of the brake is usually rotors two and three, along with their adjacent stators. Service history shows this is the location where oxidation damage tends to appear first or where failure occurs.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/08/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

Boeing has found that oxidation damage to carbon brakes starts to be an issue when the carbon material reaches over 1000 degrees Fahrenheit (538 degrees Celsius). Repeated exposure to these temperatures will result in weakening and eventual structural failure of the carbon material. During brake removals per AMM requirements, oxidation damage can be confirmed by using a rounded chisel tip (or similar tool) and making an impression on the carbon material; if the carbon material "crumbles" or can be penetrated by the rounded chisel tip, this is an indication that the carbon material has oxidation damage.

Note: The use of cooling fans is not recommended; however, if company policy allows usage of cooling fans, it is important not to use them on the carbon brakes when the brake temperature is excessively high. When the carbon rotors have a high temperature, the rotors will appear bright orange/red in color (please see attachment within this FTD article showing a photograph of this condition). Applying cooling fans when the brake temperature is high increases the risk of thermal oxidation damage. If fans will be applied, ensure the carbon brakes do not have this appearance.

Interim Action

Since thermal oxidation damage is associated with high brake energies, Boeing recommends each Operator review its internal policies and procedures associated with landing and taxiing. This review should focus on how much energy is absorbed by the brakes.

Factors that can mitigate thermal oxidation damage are:

- Use the lowest autobrake setting based on actual airplane configuration and runway condition
- Avoid early or highspeed taxiway if the landing ground distance will be short
- Use the maximum runway if possible
- Avoid disengaging the autobrakes at high ground speeds
- Maximize the use of reverse thrust
- Minimize taxi speed when permissible
- Avoid riding the brakes while the airplane is taxiing
- Avoid leaving the parking brake engaged

Boeing recommends reviewing the 737 Flight Crew Training Manual (FCTM), specifically Chapter 6 'Landing', which contains sections titled:

- Carbon Brake Life
- Brake Cooling
- Minimum Brake Heating
- Reverse Thrust Operation

Boeing would like to emphasize the importance of performing a brake assembly inspection during a standard wheel-tire assembly change per 737NG/MAX AMM 32-41-41 requirements. Please be aware during a standard wheel-tire assembly change per AMM requirements, the AMM will provide reference to the brake assembly inspection (AMM 32-41-41 section 'Wheel Removed from the Airplane') in order to ensure the brake heatsink is serviceable for continued service. The AMM brake

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/08/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

inspection contains multiple photographs of oxidation damage on carbon disks and stator alignment groove. If this condition is found during a standard wheel-tire assembly change, this brake assembly is required to be replaced before next scheduled flight. The intent of the AMM procedure is to remove a brake assembly with oxidation damage before its carbon disks fail completely.

Additional Information:

The 737MAX has a customer option known as the “Autobrake System - Revise Setting for Deceleration Rate” which adjusts the deceleration rates of the autobrake system. This customer option was created to reduce the gap in deceleration rates between the originally defined (“Baseline”) Autobrake 3 and Autobrake MAX setting, and was requested by customers who were focused on runway exit distance/runway occupancy rules. In order to accomplish the desired decelerations, and thus shorter stopping distances, the customer option allowed redefines the autobrake system targets to a higher deceleration rate than the “Baseline” Autobrake 3, creating a new definition for Autobrake 3. The autobrake switch was not changed but the resulting deceleration settings were adjusted.

Please see attached ‘Deceleration Setting’ file along with the following:

- Autobrake MAX: deceleration rate remains the same between the “Baseline” and “Optional” Autobrake settings
- Autobrake 3: The “Optional” Autobrake 3 setting is the new deceleration rate
- Autobrake 2 & 1: the “Baseline” Autobrake 3 and Autobrake 2 become the “Optional” Autobrake 2 and Autobrake 1, respectively.
- The “Baseline” Autobrake 1 setting is removed from the Optional Autobrake system

For a specified landing, the Adjusted Autobrake settings allow an airplane to utilize less runway distance to slow or stop and vacate the runway more quickly than the Baseline Autobrake settings. However, utilization of the Adjusted Autobrake settings 1-3 increase the energy absorbed by the brakes when compared to the corresponding Baseline Autobrake settings. The increased brake energy absorption will result in higher brake temperatures, and may lead to brake thermal oxidation. The use of minimum reverse thrust as compared to maximum reverse thrust can double the brake energy requirements and result in brake temperatures much higher than normal. Boeing is sharing the above information so airlines will be aware of the impact these choices can have on brake wear and potential thermal oxidation. Airline operational considerations and airport conditions will clearly drive the choice of autobrake settings, thrust reverser use, etc.

Final Action

Boeing and our brake Suppliers will continue to support each Operator as required to minimize the effects of thermal oxidation damage. If a high level of thermal oxidation is being observed, please submit a Service Request to Boeing to further work with your organization and your flight operations.

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/08/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

FLEET TEAM DIGEST

In addition, Boeing encourages each Operator to review the following documents to mitigate the effects of thermal oxidation damage.

1. The Boeing Company applicable documents:

- Fleet Team Digest (FTD) article 737MAX-FTD-32-17021 'Main Gear Brake Problems Traced to Overheated Brakes'
- Service Bulletin 737-32-1527 'Antiskid/Autobrake System - Antiskid Valves Replacement'
- Presentation: 'Brake Energy Considerations in Flight Operations', accessible through MyBoeingFleet (MBF) under App called 'Flight Operations'. Select link called: "Events, Training & Resources", select year 2003 link, followed by link called 'Presentations'.
- White Paper called: 'Brake Energy Considerations in Flight Operations', accessible through MyBoeingFleet (MBF) under App called 'Flight Operations'. Select link called: "Events, Training & Resources", select year 2003 link, followed by link called 'White Papers'.

2. If using Collins Aerospace hardware, please contact your local Collins Field Service Representative for more information regarding this topic.

3. If using Safran Landing System hardware, review applicable documents:

- Service Bulletin C20633-32-014 ‘Landing Gear – Carbon Brake Assembly’
- Presentation: ‘737-Impact of Operations’, please contact your local Safran Field Service Representative to obtain a copy.

Milestones

This FTD is now closed and no further revisions will be incorporated as of 15-Nov-2023.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737MAX-FTD-32-17021		
Service Bulletin (SB)	737-32-1527		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Brake Assy	S277A016-312	Collins Aerospace	P/N 2-1740-1

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/08/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

Brake Assy	S277A016-510	Safran Landing Systems	P/N C20633000

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- Flight Operations : Flight Operations

Attachments:

a) 737 Hot Carbon Rotors

b) Deceleration_Setting

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/08/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991